

Class 12 NEB Computer Science Notes

Chapter 1.1: Introduction to Data, Database, and DBMS

1.1 INTRODUCTION TO DATA, DATABASE, AND DBMS

Class 12 NEB Computer Science Notes

1. DATA (डाटा)

Definition (English)

Data is a collection of raw facts, figures, symbols, or observations that have not yet been processed into meaningful information.

परिभाषा (नेपाली)

डाटा भनेको कुनै पनि तथ्य, अंक, चिन्ह वा अवलोकनहरूको कच्चा रूप हो, जसलाई अझै प्रसोधन गरेर अर्थपूर्ण जानकारी बनाइएको हुँदैन।

Examples

- Name: Ram
- Age: 18
- Marks: 95
- Roll No.: 12

These are only raw values, so they are data.



INFORMATION (जानकारी)

Information is processed and organized data that has meaning and is useful for decision-making.

Example

Data	Information
95, 88, 76	Average marks = 86.3
18	Student's age is 18 years

Difference between Data and Information

Data	Information
Raw facts	Processed data
No meaning by itself	Meaningful
Input	Output
Unorganized	Organized
Used for processing	Used for decision making

Characteristics of Good Information



2. DATABASE (डाटाबेस)

Definition (English)

A database is an organized collection of related data stored electronically so that it can be easily accessed, managed, updated, and retrieved.

परिभाषा (नेपाली)

डाटाबेस भनेको सम्बन्धित डाटाहरूलाई व्यवस्थित रूपमा कम्प्युटरमा संग्रह गरेर आवश्यक पर्दा सजिलै खोज्न, परिवर्तन गर्न र प्रयोग गर्न मिल्ने व्यवस्था हो।

Examples of Database

- School Management System
- Hospital Records
- Library Management System
- Banking System
- Facebook User Database
- Instagram Database



Student Database Example

Roll	Name	Class	Marks
1	Ram	12	85
2	Sita	12	92
3	Hari	12	78

3. DATABASE MANAGEMENT SYSTEM (DBMS)

Definition (English)

A Database Management System (DBMS) is software that allows users to create, store, organize, retrieve, update, and manage databases efficiently.

परिभाषा (नेपाली)

DBMS भनेको डाटाबेस सिर्जना गर्ने, व्यवस्थापन गर्ने, सम्पादन गर्ने, खोज्ने र सुरक्षित राख्ने सफ्टवेयर हो।

Examples of DBMS



Functions of DBMS

1. Data Storage Stores data permanently.	2. Data Retrieval Retrieves required data quickly.	3. Data Update Allows modification of existing data.	4. Data Security Protects unauthorized access.
5. Backup and Recovery Recovers data after failure.	6. Data Sharing Allows multiple users to access data simultaneously.	7. Data Integrity Maintains accuracy and consistency.	8. Reduces Data Redundancy Avoids unnecessary duplication of data.

Advantages of DBMS

- ✓ Minimizes data redundancy.
- ✓ Maintains consistency.
- ✓ Better security.
- ✓ Data sharing.
- ✓ Backup and recovery.
- ✓ Faster data access.
- ✓ Data integrity.
- ✓ Multi-user support.

Disadvantages of DBMS

- ✗ Expensive software.
- ✗ Requires trained personnel.
- ✗ Complex to manage.
- ✗ Large storage requirement.
- ✗ Maintenance cost is high.

Components of DBMS



Users of DBMS

Database Administrator (DBA)	• Controls the database. • Manages security. • Takes backup.
Application Programmer	• Develops database applications.
End User	• Uses the database.

Real-Life Applications of DBMS

• Banking	• E-commerce
• Hospital Management	• Government Offices
• School/College Management	• Telecommunication
• Railway Reservation	• Social Media
• Airlines	• Online Shopping

Difference Between File System and DBMS

S.N.	File System	DBMS
1.	Stores data in files	Stores data in databases
2.	More redundancy	Less redundancy
3.	Lower security	Higher security
4.	Difficult backup	Easy backup
5.	Limited data sharing	Multi-user access
6.	Slow searching	Fast searching
7.	Poor consistency	High consistency

Important Exam Definitions (2 Marks)

- Data** : Data is a collection of raw facts and figures that have not yet been processed into meaningful information.
- Database** : A database is an organized collection of related data stored electronically for easy access, management, and retrieval.
- DBMS** : A DBMS is software used to create, store, retrieve, update, and manage databases efficiently.

Frequently Asked NEB Questions

- | | |
|---|--|
| 2 Marks Questions <ol style="list-style-type: none">1. Define data.2. Define database.3. Define DBMS.4. Write any two advantages of DBMS.5. Write any two disadvantages of DBMS.6. Differentiate between data and information.7. What is a database? Give one example.8. What is data redundancy?9. What is data integrity?10. What is a DBA? | 5 Marks Questions <ol style="list-style-type: none">1. Explain DBMS with its advantages and disadvantages.2. Differentiate between File System and DBMS.3. Explain the functions of DBMS.4. Explain the components of DBMS.5. Explain different users of DBMS.6. Explain the applications of DBMS in daily life. |
|---|--|

Quick Revision

Data → Raw facts.	Information → Processed data.	Database → Organized collection of related data.	DBMS → Software used to manage databases.	Examples of DBMS → MySQL, Oracle, SQL Server, PostgreSQL, SQLite, MS Access.	Major Advantages → Security, reduced redundancy, backup, integrity, data sharing.	Major Disadvantages → Costly, complex, requires skilled users.
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NEB Tip: Definitions, differences (especially Data vs Information and File System vs DBMS), functions, advantages/disadvantages, and applications of DBMS are among the most frequently tested topics in Class 12 Computer Science exams.

1. Data (डाटा)

Definition (English)

Data is a collection of raw facts, figures, symbols, or observations that have not yet been processed into meaningful information.

परिभाषा (नेपाली)

डाटा भनेको कुनै पनि तथ्य, अंक, चिन्ह वा अवलोकनहरूको कच्चा रूप हो, जसलाई अझै प्रशोधन गरेर अर्थपूर्ण जानकारी बनाइएको हुँदैन।

Examples

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These are only raw values, so they are **data**.

Information (जानकारी)

Information is processed and organized data that has meaning and is useful for decision-making.

Example

Data	Information
95, 88, 76	Average marks = 86.3
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Difference between Data and Information

<i>Data</i>	<i>Information</i>
<i>Raw facts</i>	Processed data
<i>No meaning by itself</i>	Meaningful
<i>Input</i>	Output
<i>Unorganized</i>	Organized
<i>Used for processing</i>	Used for decision making

Characteristics of Good Information

- Accurate
- Complete
- Relevant
- Timely
- Reliable
- Understandable

2. Database (डाटाबेस)

Definition (English)

A database is an organized collection of related data stored electronically so that it can be easily accessed, managed, updated, and retrieved.

परिभाषा (नेपाली)

डाटाबेस भनेको सम्बन्धित डाटाहरूलाई व्यवस्थित रूपमा कम्प्युटरमा संग्रह गरेर आवश्यक पर्दा सजिलै खोज्न, परिवर्तन गर्न र प्रयोग गर्न मिल्ने व्यवस्था हो।

Examples of Database

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Student Database Example

Roll	Name	Class	Marks
1	Ram	12	85
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3	Hari	12	78

Advantages of Database

- Stores large amounts of data.
- Easy to search.
- Easy to update.
- Reduces duplication.
- Improves security.
- Saves time.

3. Database Management System (DBMS)

Definition (English)

A Database Management System (DBMS) is software that allows users to create, store, organize, retrieve, update, and manage databases efficiently.

परिभाषा (नेपाली)

DBMS भनेको डाटाबेस सिर्जना गर्ने, व्यवस्थापन गर्ने, सम्पादन गर्ने, खोज्ने र सुरक्षित राख्ने सफ्टवेयर हो।

Examples of DBMS

- MySQL
 - Oracle Database
 - Microsoft SQL Server
 - PostgreSQL
 - SQLite
 - Microsoft Access
-

Functions of DBMS

1. Data Storage

Stores data permanently.

2. Data Retrieval

Retrieves required data quickly.

3. Data Update

Allows modification of existing data.

4. Data Security

Protects unauthorized access.

5. Backup and Recovery

Recovers data after failure.

6. Data Sharing

Allows multiple users to access data simultaneously.

7. Data Integrity

Maintains accuracy and consistency.

8. Reduces Data Redundancy

Avoids unnecessary duplication of data.

Advantages of DBMS

- Minimizes data redundancy.
- Maintains consistency.
- Better security.
- Data sharing.
- Backup and recovery.
- Faster data access.
- Data integrity.
- Multi-user support.

Disadvantages of DBMS

- Expensive software.
 - Requires trained personnel.
 - Complex to manage.
 - Large storage requirement.
 - Maintenance cost is high.
-

Components of DBMS

1. Hardware
 2. Software
 3. Data
 4. Users
 5. Procedures
-

Users of DBMS

Database Administrator (DBA)

- Controls the database.
- Manages security.
- Takes backup.

Application Programmer

- Develops database applications.

End User

- Uses the database.
-

Real-Life Applications of DBMS

- Banking
 - Hospital Management
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Difference Between File System and DBMS

File System	DBMS
Stores data in files	Stores data in databases
More redundancy	Less redundancy
Lower security	Higher security

Difficult backup	Easy backup
Limited data sharing	Multi-user access
Slow searching	Fast searching
Poor consistency	High consistency

Important Exam Definitions (2 Marks)

Data

Data is a collection of raw facts and figures that have not yet been processed into meaningful information.

Database

A database is an organized collection of related data stored electronically for easy access, management, and retrieval.

DBMS

A DBMS is software used to create, store, retrieve, update, and manage databases efficiently.

Frequently Asked NEB Questions

2 Marks

1. Define data.
 2. Define database.
 3. Define DBMS.
 4. Write any two advantages of DBMS.
 5. Write any two disadvantages of DBMS.
 6. Differentiate between data and information.
 7. What is a database? Give one example.
 8. What is data redundancy?
 9. What is data integrity?
 10. What is a DBA?
-

5 Marks

1. Explain DBMS with its advantages and disadvantages.
 2. Differentiate between File System and DBMS.
 3. Explain the functions of DBMS.
 4. Explain the components of DBMS.
 5. Explain different users of DBMS.
 6. Explain the applications of DBMS in daily life.
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Quick Revision

- **Data** → Raw facts.
- **Information** → Processed data.
- **Database** → Organized collection of related data.
- **DBMS** → Software used to manage databases.
- **Examples of DBMS** → MySQL, Oracle, SQL Server, PostgreSQL, SQLite, MS Access.
- **Major Advantages** → Security, reduced redundancy, backup, integrity, data sharing.
- **Major Disadvantages** → Costly, complex, requires skilled users.
- **File System vs DBMS** → DBMS provides better security, consistency, backup, and multi-user support.